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Student Name:

Sampling Words

EXPLORATION 2.1A

1. Select a representative set of 10 words from the passage by circling them with your pen or pencil.

> Four score and seven years ago, our fathers brought forth upon this continent a new nation: conceived in liberty, and dedicated to the proposition that all men are created equal.
>
> Now we are engaged in a great civil war, testing whether that nation, or any nation so conceived and so dedicated, can long endure. We are met on a great battlefield of that war.
>
> We have come to dedicate a portion of that field as a final resting place for those who here gave their lives that that nation might live. It is altogether fitting and proper that we should do this.
>
> But, in a larger sense, we cannot dedicate, we cannot consecrate, we cannot hallow this ground. The brave men, living and dead, who struggled here have consecrated it, far above our poor power to add or detract. The world will little note, nor long remember, what we say here, but it can never forget what they did here.
>
> It is for us the living, rather, to be dedicated here to the unfinished work which they who fought here have thus far so nobly advanced. It is rather for us to be here dedicated to the great task remaining before us, that from these honored dead we take increased devotion to that cause for which they gave the last full measure of devotion, that we here highly resolve that these dead shall not have died in vain, that this nation, under God, shall have a new birth of freedom, and that government of the people, by the people, for the people, shall not perish from the earth.

Definitions

The ***population*** is the entire collection of observational units we are interested in, and the ***sample*** is a subset of the population on which we record data. Numerical summaries about a population are called ***parameters***, and numerical summaries calculated from a sample are called ***statistics***.

The authorship of literary works is often a topic for debate. For example, researchers have tried to determine whether some of the works attributed to Shakespeare were actually written by Bacon or Marlow. The field of "literary computing" examines ways of numerically analyzing authors' works, looking at variables such as sentence length and rates of occurrence of specific words.

The passage is, of course, Lincoln's Gettysburg Address, given November 19, 1863, on the battlefield near Gettysburg, Pennsylvania. We are considering this passage a *population* of words, and the 10 words you selected are considered a *sample* from this population. In most studies, we do not have access to the entire population and can only consider results for a sample from that population, but to learn more about the process of sampling and its implications, we will now deal with a somewhat artificial scenario where we sample from this known population.

2. Record each word from your sample and then indicate the length of the word (number of letters) and whether or not the word contains at least one letter *e*.

Word	Length (no. of letters)	Contains *e*? (Y or N)
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		

Definition

A ***data table*** (or statistical spreadsheet) is a convenient way to store and represent all data values. Typically, in a data table, the rows correspond to observational units, columns represent variables, and the data are the table entries.

The table you filled in above is called a data table.

3. Identify the observational units and the variables you have recorded on those observational units. (Keep in mind that observational units do not have to be people!)

4. Is the variable "length of word" quantitative or categorical?

NOTE

When we are sampling from a finite population (e.g., all 268 words), we can consider the ***parameter*** to be a numerical summary of the variable in the population. With quantitative variables, we will often be interested in the population mean. With categorical variables, we will often be interested in the population proportion. To distinguish these values from the corresponding statistics, we will use Greek letters, μ for a population mean, σ for the population standard deviation, and π for a population proportion. In contrast, common symbols for sample statistics are $\bar{x}$ for sample mean, s for the sample standard deviation, and $\hat{p}$ for sample proportion.

NOTATION CHECK

Here is a quick summary of the symbols we've introduced in this section.

Type of Variable	Quantitative	Categorical
Statistics	$\bar{x}$ = sample mean	$\hat{p}$ = sample proportion
	s = sample standard deviation	
Parameters	μ = population mean	π = population proportion
	σ = population standard deviation	

5. Calculate the average length of the 10 words in your sample. Is this number a parameter or a statistic? Explain how you know. What symbol would you use to refer to this value?

6. The average length of the 268 words in the entire speech equals 4.29 letters. Is this number a parameter or a statistic? Explain how you know. What symbol would you use to refer to this value?

7. Calculate the proportion of words in your sample that contain at least one *e*. Is this number a parameter or a statistic? Explain how you know. What symbol would you use to refer to this value?

8. The proportion of all words in the entire speech that contain at least one letter *e* is 125/268 ≈ 0.47. Is this number a parameter or a statistic? Explain how you know. What symbol would you use to refer to this value?

9. Do you think the words you selected are ***representative*** of the 268 words in this passage? Suggest a method for deciding whether you have a representative sample. (*Hint:* Whereas any one sample may not produce statistics that exactly equal the population parameters, what would we like to be true in general?)

10. Combine your results with your classmates' by producing a dotplot of the distribution of average word lengths in your samples. Be sure to label the axis of this dotplot appropriately.

a. Explain what each dot on the dotplot represents by filling in the following sentence. Each dot on the dotplot is a single measurement of the value of _____ on a single _____.

b. Describe the shape, center, and variability of the distribution of average word lengths as revealed in the dotplot.

To enter special characters and formatting, use "Ctrl+E" in Windows and "Cmd + E" in Mac.

11. Let's compare your sample statistics to the population parameters.

a. How many and what proportion of students in your class obtained a sample average word length larger than 4.29 letters, the average word length in the population?

b. How many and what proportion of students in your class obtained a sample proportion of *e*-words larger than 0.47, the proportion of *e*-words in the population? (You might use a show of hands in class to answer this question.)

Note that bias is a property of a sampling *method*, not a property of an individual sample. Also note that the sampling method must *consistently* produce nonrepresentative results in the same direction in order to be considered biased.

12. What do the answers to #11 tell you about whether the sampling method of asking students to quickly pick 10 representative words is biased or unbiased? If biased, what is the direction of the bias (tendency to overestimate or to underestimate)?

Definition

A sampling method is ***biased*** if, when using that sampling method, statistics from different samples *consistently* overestimate or consistently underestimate the population parameter of interest.

13. Explain why we might have expected this sampling method (asking you to quickly pick 10 representative words) to be biased.

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14. Do you think asking each of you to take 20 words instead of 10 words would have helped with this issue? Explain.

Now consider a different sampling method: What if you were to close your eyes and point blindly at the page with a pencil 10 times, taking for your sample the 10 words on which your pencil lands.

15. Do you think this sampling method is likely to be biased? If so, in which direction? Explain.

A sample is only useful to us if the data we collect on our sample are similar to the results we would find in the entire population. In this sense, we say the sample is ***representative*** of the population.

16. Suggest another technique for selecting 10 words from this population in order for the sample to be representative of the population with regard to word length and *e*-words.

Taking a Simple Random Sample

KEY IDEA

A ***simple random sample*** ensures that every sample of size *n* is equally likely to be the sample selected from the population. In particular, each observational unit has the same chance of being selected as every other observational unit.

The key to obtaining a representative sample is using some type of *random* mechanism to select the observational units from the population, rather than relying on ***convenience samples*** or any type of human judgment. Instead of having you choose "random" words using your own judgment, we will now ask you to take a simple random sample of words and evaluate your results. The first step is to obtain a ***sampling frame***—a complete list of every member of the population where each member of the population can be assigned a number. Below is a copy of the Gettysburg Address that includes numbers in front of every word. For example, the 43rd word is *nation*.

17. Go to the **Random Numbers** applet (or http://www.random.org).

Specify that you want 5 **Numbers per replication** in the range from 1 to 268.

Press **Generate** to view the 5 random numbers. Enter the random numbers in the table below:

Generate Random Number

Number of replications: 1

Numbers per replication: 5

Number range: From: 1

To: 268

Randomly generated five ID values from 1 to 268.

Using your randomly generated values, look up the corresponding word from the sampling frame. Fill in the data table below.

Word	Length (no. of letters)	*e*-word? (Y or N)
1.		
2.		
3.		
4.		
5.		

Gettysburg address sampling frame

1	Four	11	upon	21	dedicated	31	Now	41	whether	51	dedicated
2	score	12	this	22	to	32	we	42	that	52	can
3	and	13	continent	23	the	33	are	43	nation	53	long
4	seven	14	a	24	proposition	34	engaged	44	or	54	endure
5	years	15	new	25	that	35	in	45	any	55	We
6	ago	16	nation	26	all	36	a	46	nation	56	are
7	our	17	conceived	27	men	37	great	47	so	57	met
8	fathers	18	in	28	are	38	civil	48	conceived	58	on
9	brought	19	liberty	29	created	39	war	49	and	59	a
10	forth	20	and	30	equal	40	testing	50	so	60	great

61	battlefield	96	and	131	far	166	living	201	before	236	died
62	of	97	proper	132	above	167	rather	202	us	237	in
63	that	98	that	133	our	168	to	203	that	238	vain
64	war	99	we	134	poor	169	be	204	from	239	that
65	We	100	should	135	power	170	dedicated	205	these	240	this
66	have	101	do	136	to	171	here	206	honored	241	nation
67	come	102	this	137	add	172	to	207	dead	242	under
68	To	103	But	138	or	173	the	208	we	243	God
69	dedicate	104	in	139	detract	174	unfinished	209	take	244	shall
70	a	105	a	140	The	175	work	210	increased	245	have
71	portion	106	larger	141	world	176	which	211	devotion	246	a
72	of	107	sense	142	will	177	they	212	to	247	new
73	that	108	we	143	little	178	who	213	that	248	birth
74	field	109	cannot	144	note	179	fought	214	cause	249	of
75	as	110	dedicate	145	nor	180	here	215	for	250	freedom
76	a	111	we	146	long	181	have	216	which	251	and
77	final	112	cannot	147	remember	182	thus	217	they	252	that
78	resting	113	consecrate	148	what	183	far	218	gave	253	government
79	place	114	we	149	we	184	so	219	the	254	of
80	for	115	cannot	150	say	185	nobly	220	last	255	the
81	those	116	hallow	151	here	186	advanced	221	full	256	people
82	who	117	this	152	but	187	It	222	measure	257	by
83	here	118	ground	153	it	188	is	223	of	258	the
84	gave	119	The	154	can	189	rather	224	devotion	259	people
85	their	120	brave	155	never	190	for	225	that	260	for
86	lives	121	men	156	forget	191	us	226	we	261	the
87	that	122	living	157	what	192	to	227	here	262	people
88	that	123	and	158	they	193	be	228	highly	263	shall
89	nation	124	dead	159	did	194	here	229	resolve	264	not
90	might	125	who	160	here	195	dedicated	230	that	265	perish
91	live	126	struggled	161	It	196	to	231	these	266	from
92	It	127	here	162	is	197	the	232	dead	267	the
93	Is	128	have	163	for	198	great	233	shall	268	earth
94	altogether	129	consecrated	164	us	199	task	234	not		
95	fitting	130	It	165	the	200	remaining	235	have		

18. Let's examine your sample and those of your classmates.

a. Calculate the average word length and proportion of *e*-words in your random sample.

b. Again produce a dotplot of the distribution of sample average word lengths for yourself and your classmates.

c. Comment on how this distribution compares to the one from #10 based on nonrandom sampling.

Notice that each simple random sample gave us different values for the statistics. That is, there is variability from sample to sample (***sampling variability***). This makes sense because the samples are chosen randomly and consist of different words each time, so we expect some differences from sample to sample.

19. Let's compare your sample statistics to the population parameters.

a. How many and what proportion of students in your class obtained a sample average word length larger than the population average (4.29 letters)?

b. How many and what proportion obtained a sample proportion of *e*-words larger than the population proportion (0.47)? (You can use a show of hands.)

c. What do these answers reveal about whether simple random sampling is an unbiased sampling method? Explain.

To enter special characters and formatting, use "Ctrl+E" in Windows and "Cmd + E" in Mac.

You should see that the random samples do a better job of centering around the population parameter value than self-selected samples, *even though* we used a smaller sample size! Thus, simple random sampling would be considered an ***unbiased*** sampling method because there is no tendency to over- or underestimate the parameter. Whether a sampling method is biased or unbiased depends both on the sampling method and on the statistic you chose to calculate from the sample. Luckily, one can show that with simple random sampling the sample average and the sample proportion both have this unbiased property.

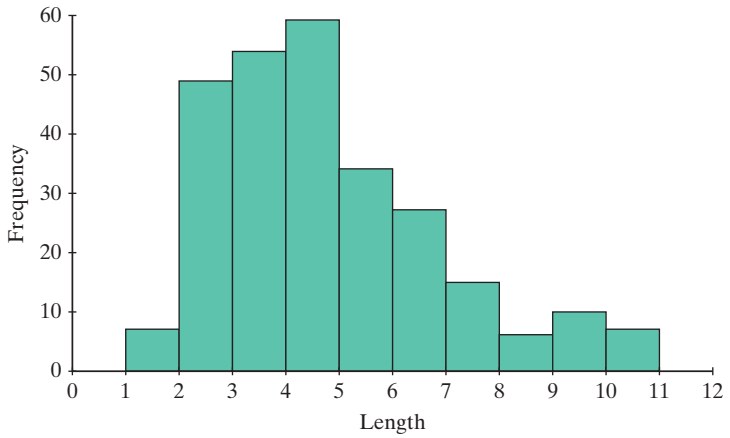
KEY IDEA

When we use ***simple random sampling***, the average of the sample means from different samples equals the population mean and the average of the sample proportions from different samples equals the population proportion. (This means that these statistics are ***unbiased*** estimators for the corresponding parameter when we use simple random sampling.)

Although selecting a simple random sample is not always feasible (e.g., what if you don't have a complete sampling frame, what if it is too difficult to track down everyone selected?), it is an example of a broader class of probability sampling methods (e.g., stratified sampling, cluster sampling; see the Stratified and Cluster Sampling Appendix) that also have the same properties. The key is that probability, not human judgment, is responsible for selecting the sample.

To really see the long-run pattern in these results, let's look at many more random samples from this population.

20. Open the **Sampling Words** applet. You will see three variables measured on all words in the Gettysburg Address: the lengths of the words, whether or not a word contains at least one letter *e*, and whether or not the word is a noun. For now, use the pull-down menu to select the length variable. Before we draw random samples from this population, we want to point out that we often use dotplots to represent a set of quantitative values. Another option is a ***histogram*** like the one shown below.



Definition

A ***histogram*** is a graph that summarizes the distribution of a quantitative variable. Bars are used to represent the number of observational units in a particular interval of values.

Like a dotplot, a histogram displays the distribution of a quantitative variable. But when you have a large number of observations like this, sometimes it may be convenient to "bin" together similar observations into one bar. The height of each bar tells you how many of the values are in each bin (interval of values). The graph still helps us see features of the distribution, as that the distribution of word lengths is not symmetric but clusters around four to five letters with some shorter words (0-4 letters) but fewer and fewer longer words (9-11 letters).

21. Check the **Show Sampling Options** box. Specify 5 in the **Sample Size** box and press the **Draw Samples** button. You will see the five selected words appear in the box below. You will also see five blue dots within the population distribution representing the five lengths that you have sampled. These five lengths are also displayed in a dotplot in the middle panel. The blue triangle indicates the value of this sample's mean, which is then added to the graph on the right. Press **Draw Samples** again. Did you get the same sample this time? Did you get the same sample mean?

You should now see two boxes in the bottom right graph, one for each sample mean (most recent is blue). The mean of these two sample means is displayed in the upper left corner of this panel. Confirm this calculation.

First sample mean:	Second sample mean:	Average:

22. Press the **Draw Samples** button three more times (for a total of five samples). Notice how the samples vary and how the sample mean varies from sample to sample. Now what is the average of your five sample means?

Change the **Number of Samples** to 995 (for a total of 1,000 samples). Press the **Draw Samples** button. (*This should generate a dotplot of 1,000 values of the sample mean for 1,000 different simple random samples, each consisting of 5 words, selected randomly from the population of all 268 words in the Gettysburg Address.)*

23. Describe the resulting distribution of sample means: What are roughly the largest and smallest values? What is the average of the distribution? What is the standard deviation? How does the average of the 1,000 sample means compare to the population mean $\mu = 4.29$?

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24. Now change the **Sample Size** from 5 to 20 and the **Number** of **Samples** to 1000. Press the **Draw Samples** button. How do these two distributions compare? Be sure to identify a feature that is similar and a feature that is different.

25. Did changing the sample size change the center of the distribution? If we used a biased sampling method, would increasing the sample size remove the bias? Explain.